

# DRAWDOWN

## THE WORLD'S LEADING RESOURCE FOR CLIMATE SOLUTIONS

Our mission is to help the world reach “Drawdown”— the point in the future when levels of greenhouse gases in the atmosphere stop climbing and start to steadily decline, thereby stopping catastrophic climate change — as quickly, safely, and equitably as possible.



Ocean Power



Our latest publication  
“Farming Our Way Out

# CLIMATE SOLUTIONS BY SECTOR

*Within each of these sectors are solutions to climate change with actions that can be taken today.*

## ELECTRICITY



## FOOD, AGRICULTURE, AND LAND USE



## INDUSTRY



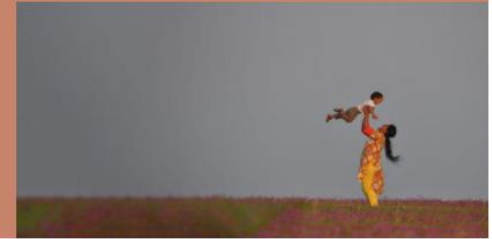
## TRANSPORTATION



## BUILDINGS



## HEALTH AND EDUCATION



## LAND SINKS



## COASTAL AND OCEAN SINKS



## ENGINEERED SINKS





Credit: yangna (iStock)

# NUCLEAR POWER

REDUCE SOURCES > Electricity > Shift Production

**2.65–3.23**

GIGATONS

CO2 EQUIVALENT  
REDUCED / SEQUESTERED  
(2020–2050)

**\$192.3**

BILLION \$US

NET FIRST COST  
(TO IMPLEMENT SOLUTION)

**\$345.1**

BILLION \$US

LIFETIME NET  
OPERATIONAL SAVINGS

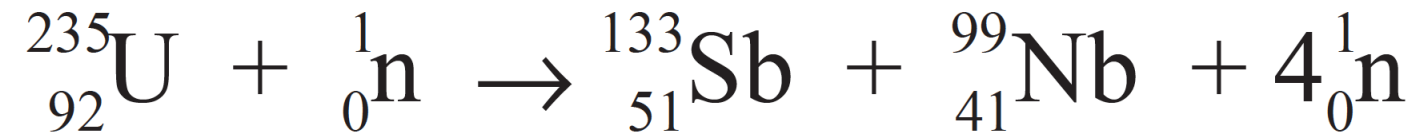
Nuclear power is slow, expensive, risky, and creates radioactive waste, but it has the potential to avoid emissions from fossil fuel electricity.

## SOLUTION SUMMARY\*

Nuclear plants use fission to split atomic nuclei and release the energy that binds protons and neutrons together. It is the most complex process ever invented to boil water, which powers steam turbines that generate electricity. Greenhouse gas emissions are calculated to be ten to a hundred times higher for coal-fired plants than for nuclear.

Currently, 29 countries have operative nuclear plants; they produce about 11 percent of the world's electricity. Nuclear is expensive, and the highly-regulated industry is often over-budget and slow. While the cost

Nuclear fission: an atom is bombarded with neutrons, causing it to break apart into other atoms + neutrons



Lots of heat (and no CO<sub>2</sub>) is produced by this...boil water, make steam, turn turbines, generate electricity

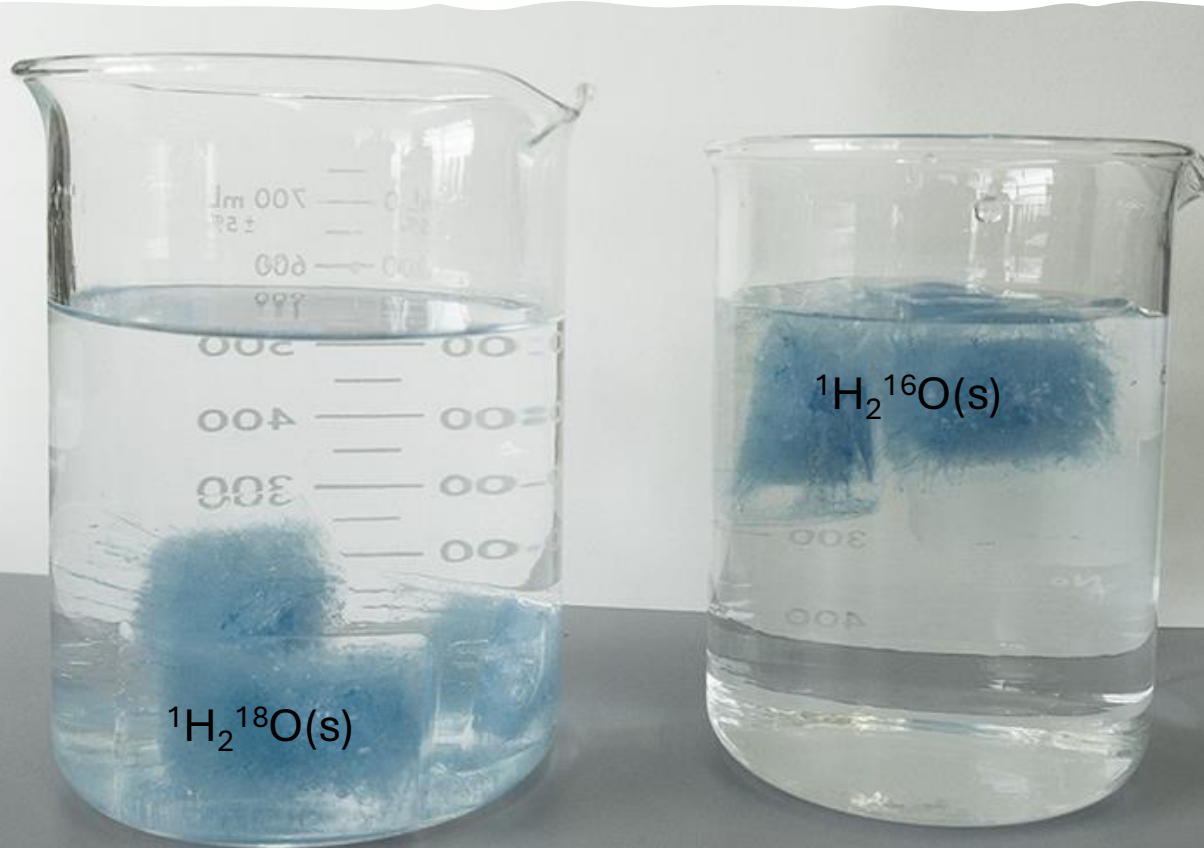
99.8% of O atoms are  $^{16}\text{O}$ ; most of the rest are  $^{18}\text{O}$

→ Most  $\text{H}_2\text{O}$  contains  $^{16}\text{O}$  rather than  $^{18}\text{O}$ , but water with  $^{18}\text{O}$  is “heavy”

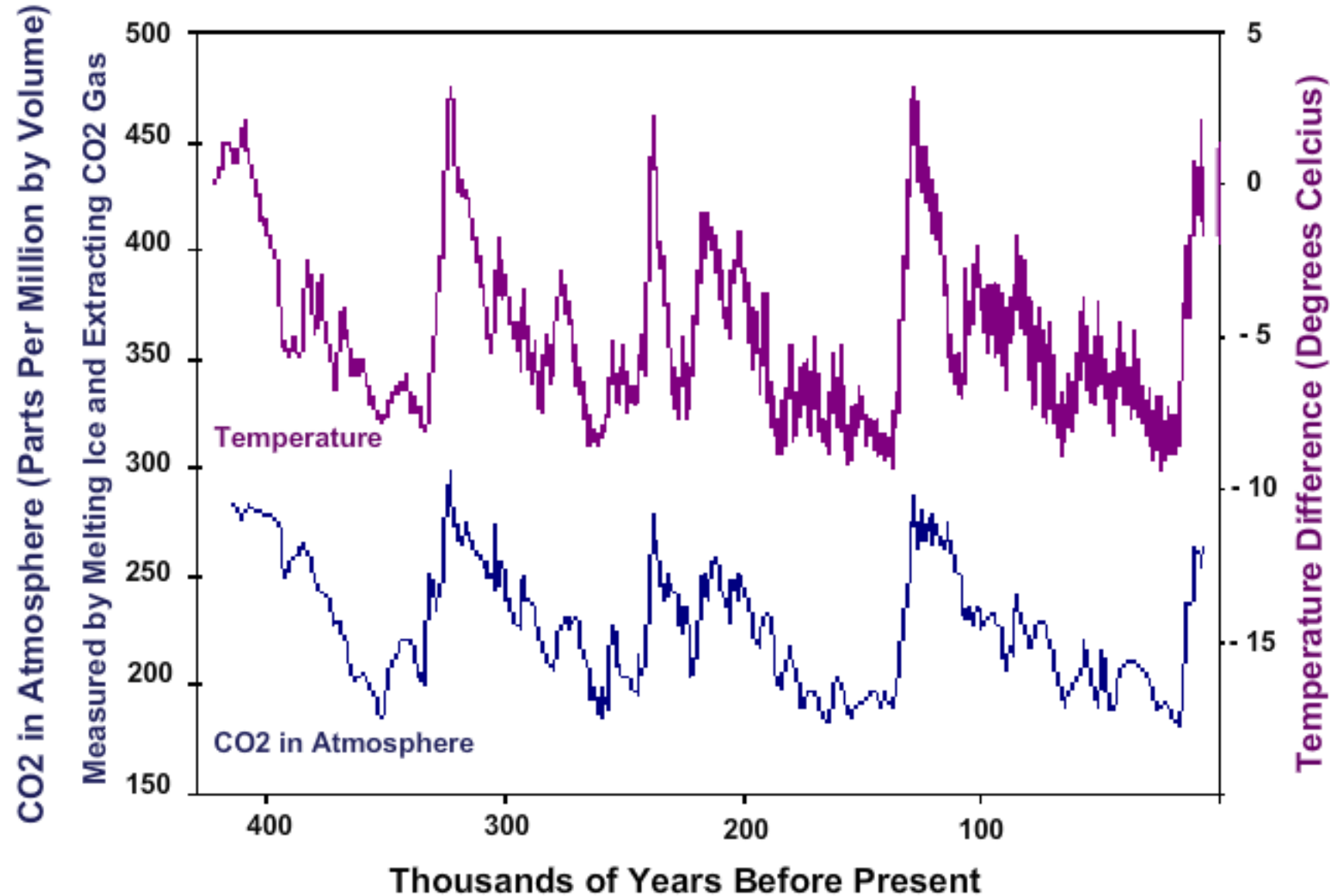
→ Heavy water is less likely to evaporate and travel to the poles [and get trapped in the ice], but in hotter climates, more heavy water makes it to the poles

Measuring the ratio of  $^{18}\text{O} / ^{16}\text{O}$  in ice cores:

→ Higher ratio means hotter temperatures



## Data Extracted from Ice Cores at the Vostok Research Station, Antarctica



Inferred from  $^{18}\text{O} / ^{16}\text{O}$  ratios in ice

→ This is how we get paleoclimate data to compare to current temperatures and  $\text{CO}_2$  levels and determine how unusual our current warming trend is.

$\text{CO}_2$  and temperature data from the Vostok Icecore, Antarctica.

Data source: Petit, J.R. et al, 2001



# ALTERNATIVE CEMENT

REDUCE SOURCES > Industry > Improve Materials

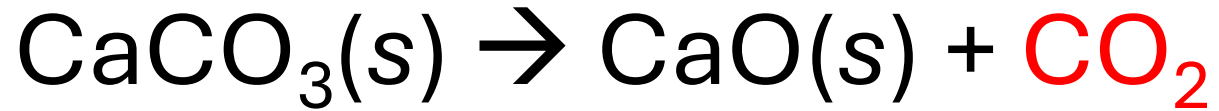
**7.98–16.1**  
GIGATONS  
CO2 EQUIVALENT  
REDUCED / SEQUESTERED  
(2020–2050)

**\$-63.46**  
BILLION \$US  
NET FIRST COST  
(TO IMPLEMENT SOLUTION)

Drawdown  
solution

# Alternative Cement

Regular cement production:



Alternative cement: replace up to half of the cement material with volcanic ash, certain clays, ground glass, waste from manufacturing iron, or waste from burning coal

→ Above CO<sub>2</sub> emissions averted!





# NUTRIENT MANAGEMENT

REDUCE SOURCES > Food, Agriculture, and Land Use > Shift Agriculture Practices

2.34–12.06

GIGATONS  
CO2 EQUIVALENT  
REDUCED / SEQUESTERED  
(2020–2050)

\$0

BILLION \$US  
NET FIRST COST  
(TO IMPLEMENT SOLUTION)

\$23.03–70.79

BILLION \$US  
LIFETIME NET  
OPERATIONAL SAVINGS

Drawdown  
solution

- Issue: soil bacteria convert nitrate ( $\text{NO}_3^-$ ) fertilizers into nitrous oxide ( $\text{N}_2\text{O}$ )—298 times more powerful than carbon dioxide in its warming effect.

→ Change fertilizer applications to be the:

- *Right source*: matching fertilizer choices with plant needs.
  - % weight of N, P, K components of fertilizer and limiting reactant for the chosen crop
- *Right time and right place*: managing fertilizer applications to deliver nitrogen when and where crop demand is highest.
- *Right rate*: ending over-application of fertilizer as “insurance.”



## REFRIGERANT MANAGEMENT

REDUCE SOURCES > ■ Industry > *Address Refrigerants*

REDUCE SOURCES > ■ Buildings > *Address Refrigerants*

**57.75**

GIGATONS

CO2 EQUIVALENT  
REDUCED / SEQUESTERED  
(2020–2050)

**\$-629.25**

BILLION \$US

LIFETIME NET  
OPERATIONAL SAVINGS

**Drawdown  
solution**

# Current refrigerants: HFCs = hydrofluorocarbons

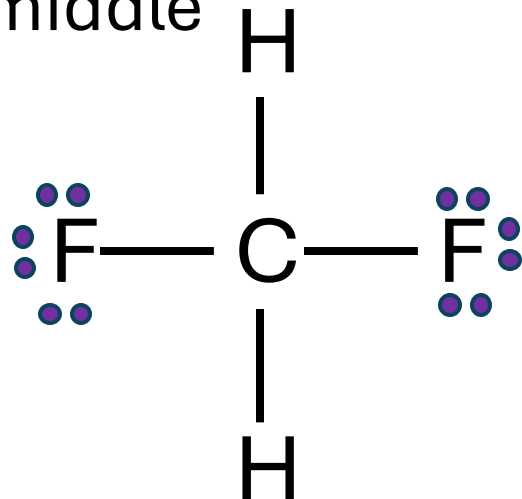
Ex:  $\text{CH}_2\text{F}_2$

1) Count valence electrons

$$4 + 2(1) + 2(7) = 20 \text{ v.e.}$$

2) Skeleton structure

C has low AVEE  $\rightarrow$  put in middle



3) Determine non-bonding e-s

$\rightarrow 4(2) = 8$  bonding e-; have

12 left

4) Check for octet rule

HFCs are ~1000x more potent greenhouse gases than  $\text{CO}_2$

$\rightarrow$  Dispose of appliances with refrigerants properly; reuse or destroy the HFCs



# ALTERNATIVE REFRIGERANTS

REDUCE SOURCES > ■ Industry > *Address Refrigerants*  
REDUCE SOURCES > ■ Buildings > *Address Refrigerants*

43.53–50.53  
GIGATONS  
CO2 EQUIVALENT  
REDUCED / SEQUESTERED  
(2020–2050)

Credit: Annie Spratt via Unsplash

Drawdown  
solution

# Alternative refrigerants

- Replace HFCs with  $\text{NH}_3$ ,  $\text{CO}_2$ ,  $\text{C}_3\text{H}_8$  → all are much less potent greenhouse gases
- Could replace ~70% of all HFC refrigerants by 2050

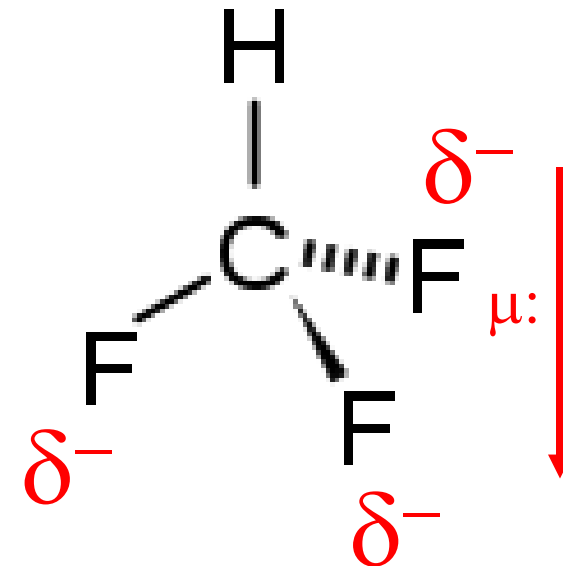
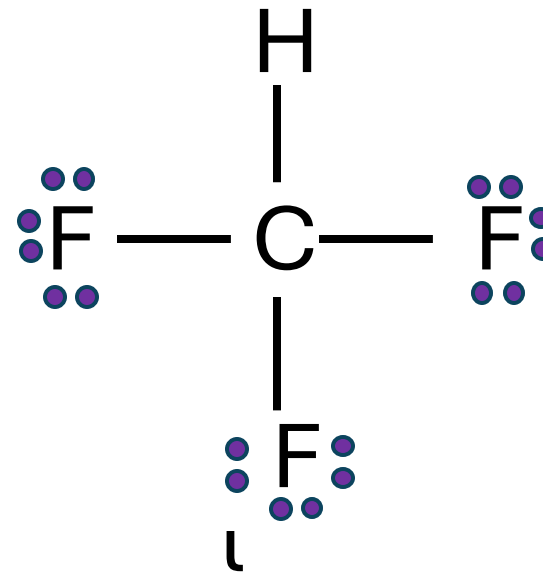
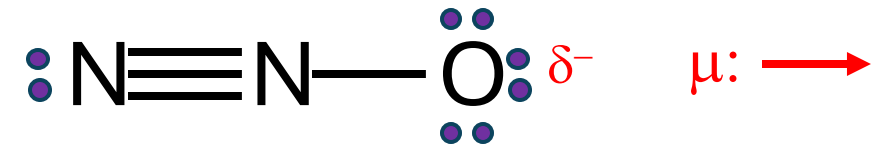
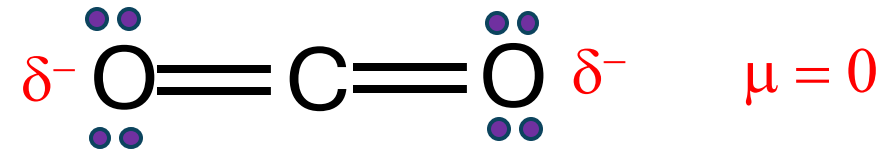
# Global Warming Potential (GWP)

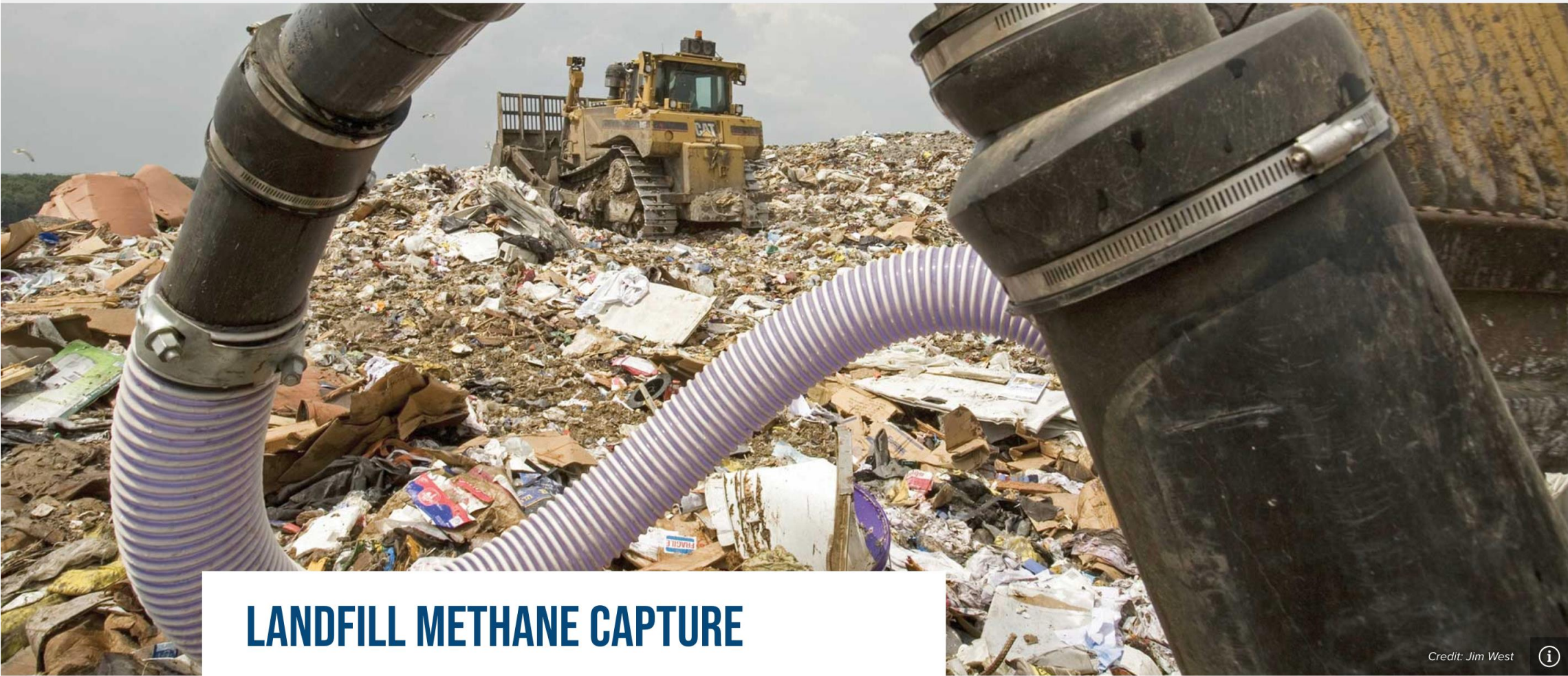
- A measure of how much heat a gas will absorb over a given time period **RELATIVE** to the amount that the same amount of  $\text{CO}_2$  would absorb
- $\text{CO}_2$ , by definition, has a GWP of 1
- $\text{CH}_4$  has a GWP of 28 over a 100-year time period, so the same amount of  $\text{CH}_4$  causes 28 times as much global warming as  $\text{CO}_2$

# Global Warming Potentials and Molecule Polarity

| Gas                | Global Warming Potential (100 year time scale) |
|--------------------|------------------------------------------------|
| CO <sub>2</sub>    | 1                                              |
| N <sub>2</sub> O   | 265                                            |
| CCl <sub>3</sub> F | 4,660                                          |
| CHF <sub>3</sub>   | 12,400                                         |

Increasing molecule polarity





# LANDFILL METHANE CAPTURE

REDUCE SOURCES > Electricity > Shift Production  
REDUCE SOURCES > Industry > Use Waste

Credit: Jim West ⓘ

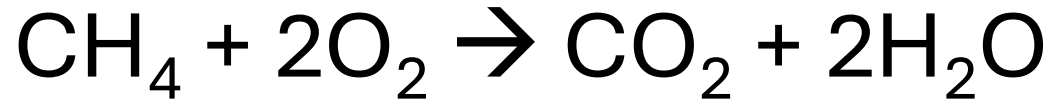
**-1.6–2.18**  
GIGATONS  
CO2 EQUIVALENT  
REDUCED / SEQUESTERED  
(2020–2050)

**\$-4.31**  
BILLION \$US  
NET FIRST COST  
(TO IMPLEMENT SOLUTION)

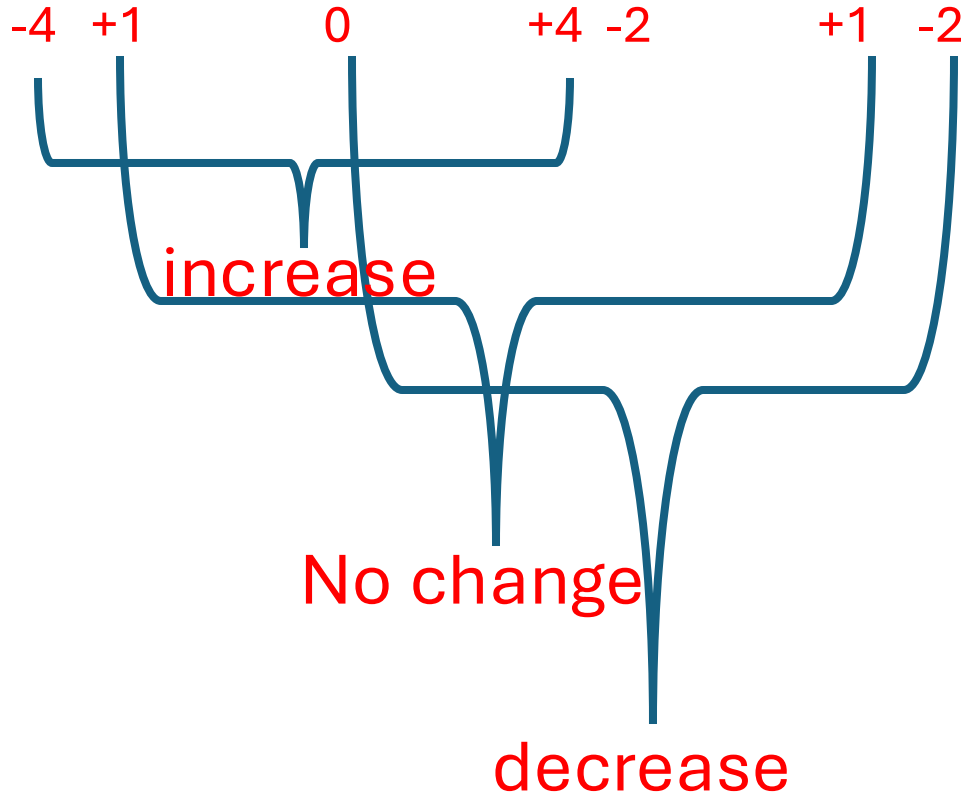
**\$-21.55–6.89**  
BILLION \$US  
LIFETIME NET  
OPERATIONAL SAVINGS

Drawdown  
solution

# Combustion is a Redox Reaction



ox #:



→ C (or  $\text{CH}_4$ ) is *oxidized* and O (or  $\text{O}_2$ ) is *reduced*.

## Climate solution:

- a) Prevent  $\text{CH}_4$  (a strong greenhouse gas) from leaking into the atmosphere; react it to make  $\text{CO}_2$ , a weaker greenhouse gas
- b) Can get electricity /heat from burning  $\text{CH}_4$  (natural gas) from this waste source



# SOLAR HOT WATER

REDUCE SOURCES > Electricity > Enhance Efficiency  
REDUCE SOURCES > Buildings > Shift Energy Sources

3.59–14.29

GIGATONS  
CO2 EQUIVALENT  
REDUCED / SEQUESTERED  
(2020–2050)

\$0.72–2.67

TRILLION \$US  
NET FIRST COST  
(TO IMPLEMENT SOLUTION)

\$0.29–1.14

TRILLION \$US  
LIFETIME NET  
OPERATIONAL SAVINGS

Drawdown  
solution

# Solar Hot Water

- $\Delta H = q = sm\Delta T$
- Cold water in, hot water out → save the energy/fossil fuels needed to heat water for showers, laundry, and sometimes building heating
  - Hot water comprises 25% of home energy use worldwide
  - Solar hot water can reduce this by 50-70%



# GREEN AND COOL ROOFS

REDUCE SOURCES > Electricity > Enhance Efficiency  
REDUCE SOURCES > Buildings > Enhance Efficiency

Credit: Arlington County

0.6–1.1  
GIGATONS  
CO2 EQUIVALENT  
REDUCED / SEQUESTERED  
(2020–2050)

\$623.99–954.72  
BILLION \$US  
NET FIRST COST  
(TO IMPLEMENT SOLUTION)

\$330.39–592.81  
BILLION \$US  
LIFETIME NET  
OPERATIONAL SAVINGS

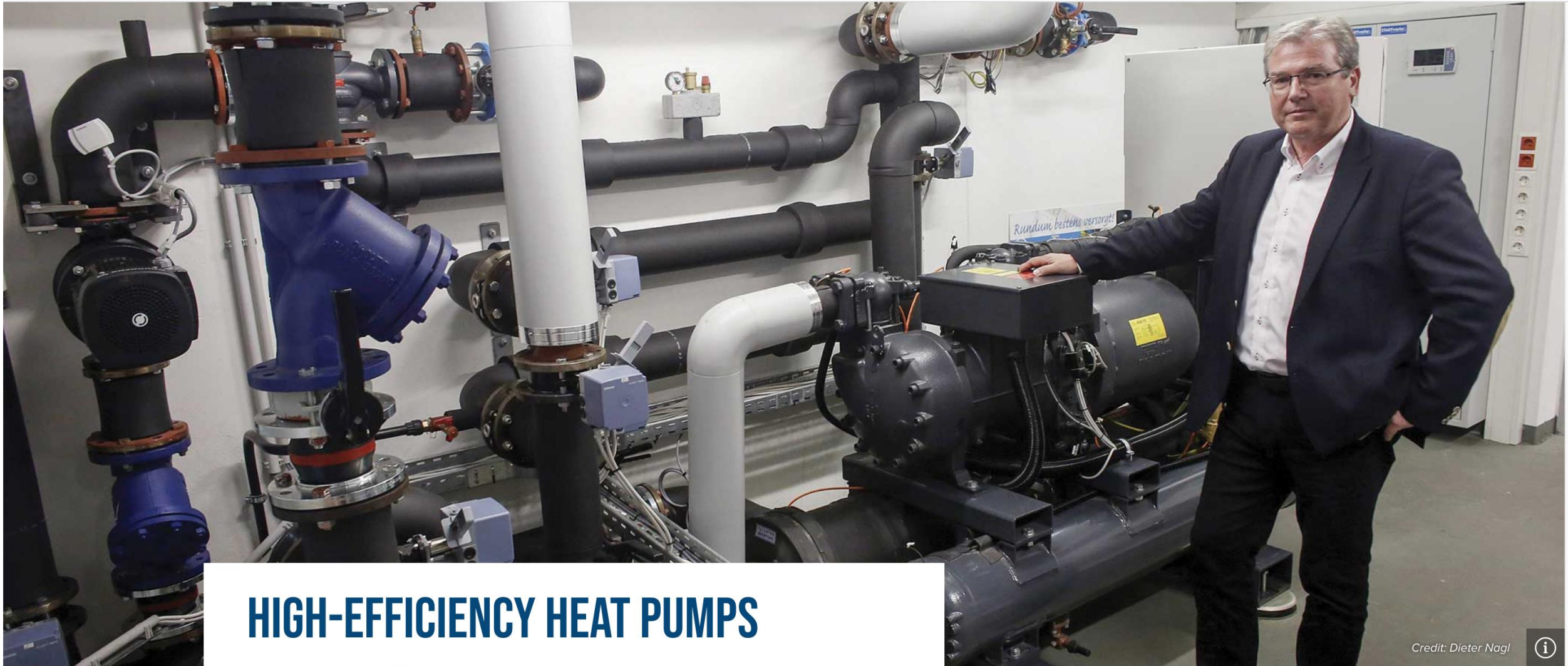
Drawdown  
solution

## Green Roofs

- All year: provide insulation to minimize heat transfer between the building and the outside air
- Summer: Soak up rainfall, when moisture evaporates the very  $\Delta H_{\text{vap}}$  of water cools the building

## Cool Roofs (for climates using more AC than heat)

- Painted white → reflect much more sunlight so it doesn't warm the building as much



Credit: Dieter Nagl



# HIGH-EFFICIENCY HEAT PUMPS

REDUCE SOURCES > Electricity > Enhance Efficiency  
REDUCE SOURCES > Buildings > Shift Energy Sources

4.16–9.29

GIGATONS  
CO2 EQUIVALENT  
REDUCED / SEQUESTERED  
(2020–2050)

\$76.67–116.82

BILLION \$US  
NET FIRST COST  
(TO IMPLEMENT SOLUTION)

\$1.09–2.50

TRILLION \$US  
LIFETIME NET  
OPERATIONAL SAVINGS

Drawdown  
solution

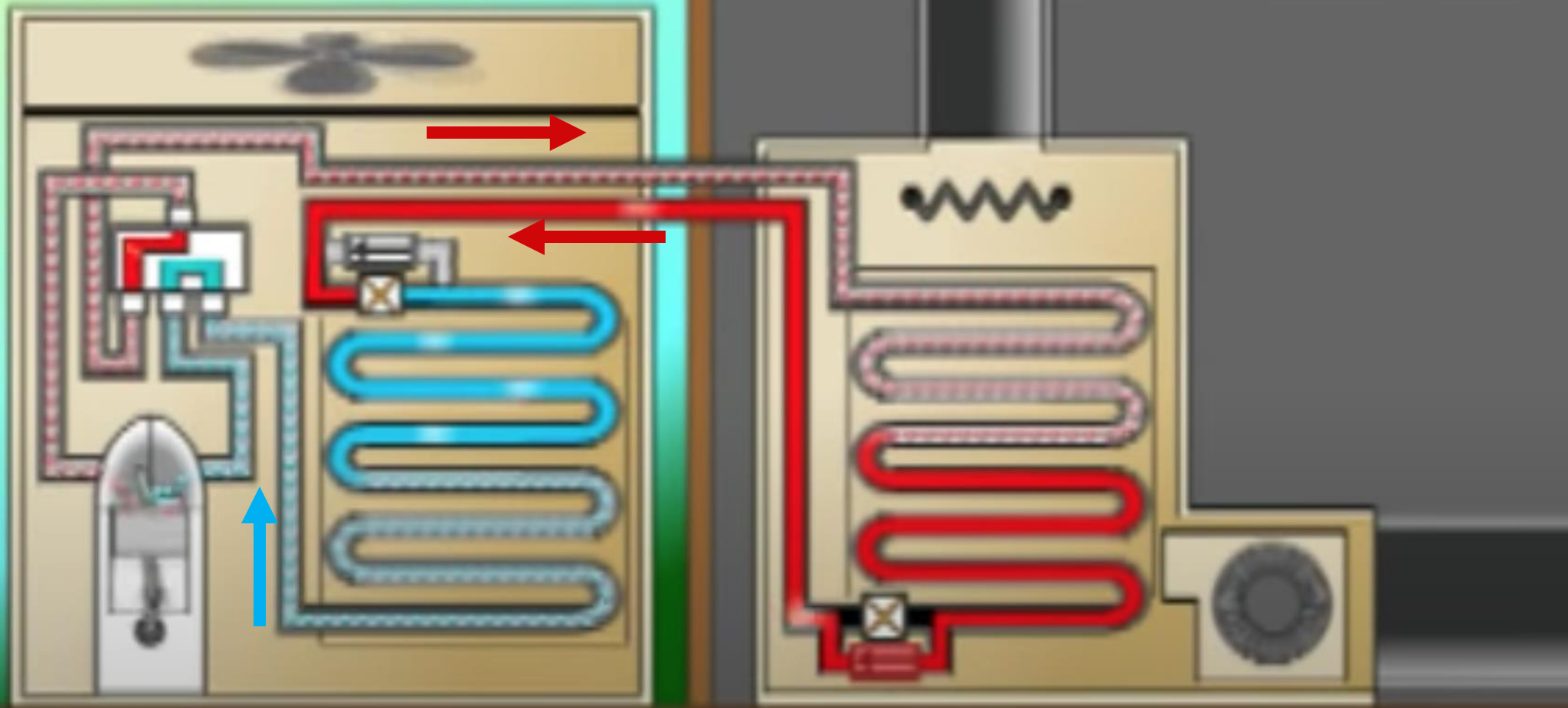
# Heat pumps

- Worldwide, buildings use ~32% of all energy generated
  - >1/3 of that energy is for heating or cooling
- Increased efficiency in heating and cooling could cut this energy use by up to 30-40% and replace fossil fuel-powered heat and lower efficiency air conditioners
- **Heat pumps transfer heat between the inside and outside of a building (using a small amount of energy)**

# Winter mode

outside

inside

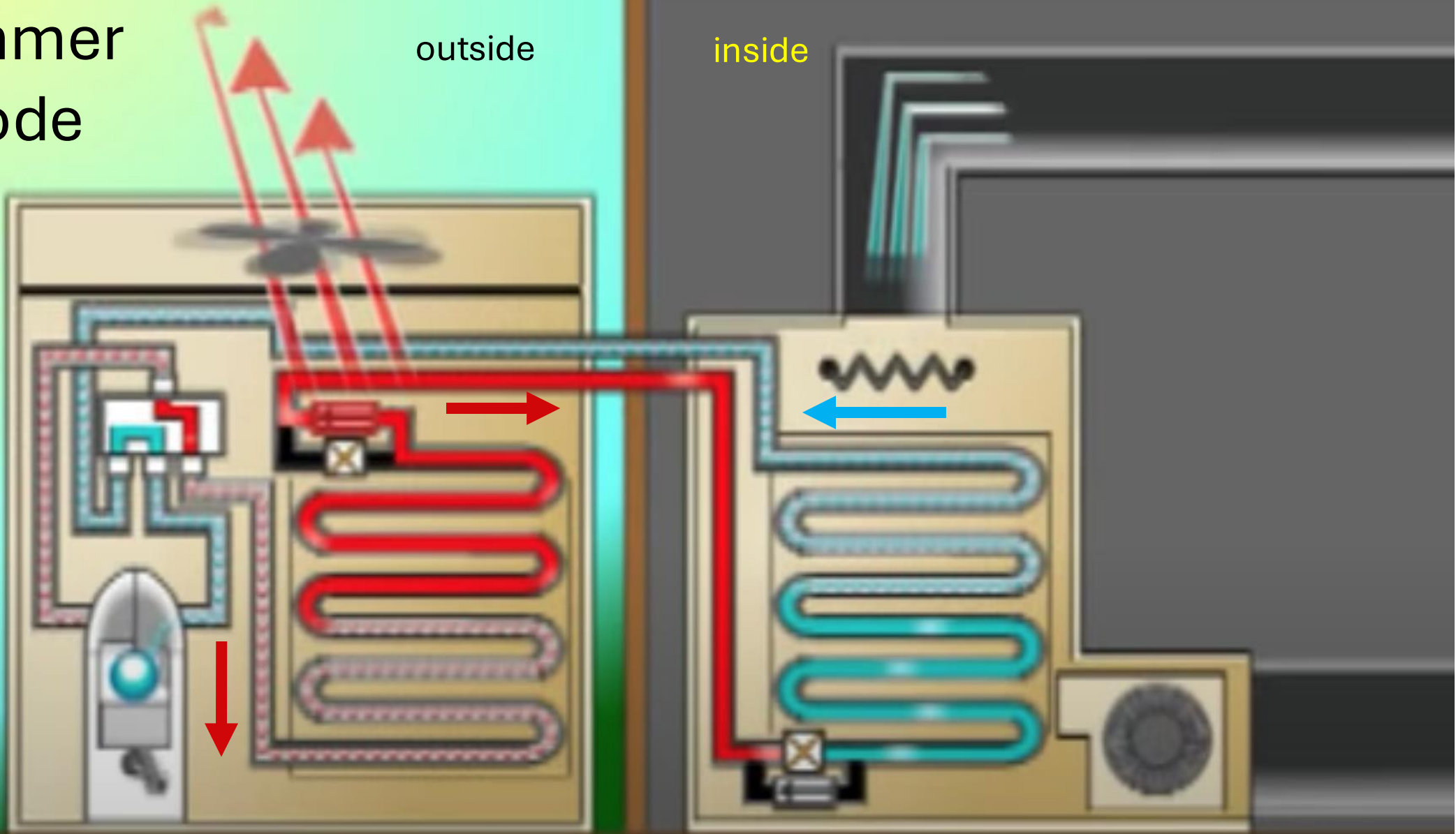


Low P → liquid evaporates

High P → gas condenses

Evaporation:  $\Delta H > 0$  (accept heat from outside)      Condensation:  $\Delta H < 0$ , (release heat to inside air)

# Summer mode



High  $P \rightarrow$  gas condenses

Low  $P \rightarrow$  liquid evaporates

Exothermic,  $\Delta H < 0$ , release heat to outside air

Endothermic,  $\Delta H > 0$ , accept heat from inside air